Europa Clipper Mission

Trajectory Design. Radiation Challenges

Meera Kothari, Mariia Sibrina, Morgan Mooso, Mounapriya Venkatesan Jayasankari

Department of Physics and Astronomy, Rice University

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Instructor: Dr. Denizhan Yavas

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Part 1: Mission Overview and Scientific Goals

Europa, along with Jupiter's three other large moons (Ganymede, Callisto, and Io), was discovered back in January 1610 by the Italian astronomer Galileo Galilei. Previously, six robotic spacecraft have explored Europa, with most of the knowledge coming from three of them: NASA's Voyager 2 spacecraft, the Galileo orbiter, and the Hubble Space Telescope. The beginning of Europa exploration takes place approximately 50 years ago. NASA's Pioneer 10 and 11 spacecraft passed through the Jovian system in the early 1970s, providing only distant images of Europa. In 1979, the twin Voyager spacecraft flew through the Jovian system, mapping the surfaces of the Galilean moons and providing more precise measurements of the moons' sizes. Voyager 2 images revealed a smooth surface marked with fracture-like linear features, fuzzy-looking "mottled terrain," and relatively few craters, suggesting something was resurfacing Europa's icy crust. Those images revealed surface ridges and cracks, hinting that it might be geologically active today. The Galileo mission provided closer views of the icy moon and found the strongest evidence of an ocean there. In 2018, scientists revisited old Galileo data and found evidence of Europa venting plumes of water vapor into space. The Hubble Space Telescope has also found intriguing evidence of plumes at Europa. Future missions include NASA's Europa Clipper, which is discussed in this report, and ESA's JUICE mission, which will explore Jupiter and its three largest icy moons, Europa, Ganymede, and Callisto (*History | Mission – NASA's Europa Clipper, n.d.*).

NASA launched the Europa Clipper spacecraft on a SpaceX Falcon Heavy rocket from Launch Complex 39A at NASA's Kennedy Space Center in Florida on Monday, October 14, 2024 (*Cowing, 2024*). It will arrive at Jupiter in 2030 and will orbit it to study its target for a few years. The spacecraft will periodically dive through Jupiter's radiation, fly by Europa to collect data, and then move away again (*Society, 2024*). The problem with the Jovian radiation will be thoroughly described further in this report.

The Europa Clipper instruments are contained within a radiation vault to safeguard them from damage. The spacecraft spans 30.5 meters and is equipped with various scientific instruments to analyze Europa from above and directly sample the moon's atmosphere. An ice-penetrating radar instrument will identify Europa's ice and potential lakes, while the Europa Clipper Magnetometer (ECM) and Plasma Instrument for Magnetic Sounding (PIMS) will assess the moon's magnetic characteristics to undoubtedly confirm the presence of the subsurface ocean. Furthermore, they will ascertain the thickness of Europa's icy crust and the underlying ocean. Two camera systems, the Europa Thermal Emission Imaging System

(E-THEMIS) and the Europa Imaging System (EIS), will survey the moon's surface and investigate for plumes. The Surface Dust Analyzer (SUDA) will search for tiny particles ejected by Europa that may indicate potential plumes. Three spectrometers (the Europa Ultraviolet Spectrograph – Europa-UVS, the Mapping Imaging Spectrometer for Europa – MISE, and the Mass Spectrometer for Planetary Exploration – MASPEX) will analyze Europa's surface and atmospheric composition to elucidate the characteristics of its concealed ocean. Alongside this equipment, the spacecraft will incorporate a propulsion module and a high-gain antenna. The radio antennas will establish communication with Earth and fulfill the mission's scientific objectives. Minor alterations in radio broadcasts induced by the Doppler Effect will enable scientists on Earth to assess the spacecraft's trajectory and get insights into Europa's composition and gravitational field. Simultaneously, Europa Clipper will survey potential landing sites for future space missions. A proposed NASA mission would involve a month-long exploration of Europa's surface, collecting samples for analysis by sensors aimed at identifying organic molecules (*Society, 2024*).

The primary scientific objectives of the Europa Clipper Mission are to identify the thickness of the moon's icy shell and its interactions with the underlying ocean, to analyze its composition, and to determine its geological features. The mission's comprehensive investigation of Europa will enhance scientists' understanding of the astrobiological potential for habitable places beyond Earth. It is believed that Europa’s massive subsurface ocean could host life. The salty ocean, estimated to contain over twice the amount of water on Earth, could be locked under the moon's icy shell, which may stretch from 3 kilometers wide to over 30 kilometers. Although sunlight cannot reach through the shell to power life, vents that release heat from the moon's interior could exist on its ocean floor, similar to vents at the bottom of Earth's oceans. Europa's ocean might get fed from both above and below. At the ocean floor, the moon's rocky interior could slowly expose new material to any life present, while the icy shell could transport molecules on its surface down toward its ocean. It is also possible that Europa launches plumes of water vapor out into space and may have lakes within its icy crust that could host life (*Society, 2024*). Thus, thorough exploration of Europa becomes a task of high significance.

The further sections will discuss the Europa Clipper Trajectory Design as well as the issues caused by the exposure to Jovian radiation.

Part 2: Trajectory Design and Flyby Mechanics

2.1. Direct Trajectory to Jupiter

A direct transfer from Earth to Jupiter entails launching the Europa Clipper onto a heliocentric trajectory without any planetary flybys. Conceptually, this approach offers reduced complexity, with a straightforward cruise phase and fewer mission-critical operations. The travel duration would be relatively short around 2.5 to 2.7 years resulting in lower navigational overhead and operational risk (*Lam et al., 2015*).

However, this trajectory demands a high characteristic energy (C3 ≈ 82 km²/s²), achievable only with heavy-lift rockets like NASA's Space Launch System (SLS) Block 1B. These vehicles are not only costly but are also heavily reserved for Artemis program missions. Additionally, the spacecraft would approach Jupiter at high speed, necessitating a substantial retrograde burn for Jupiter Orbit Insertion (JOI). This maneuver would consume excessive onboard propellant, thereby reducing payload capacity for science instruments and radiation shielding critical for survival in Jupiter's intense environment(*Lam et al., 2015*).

2.2. Gravity-Assist Trajectory Options and MEGA Selection

To overcome the challenges of a direct Earth-to-Jupiter transfer, mission planners evaluated interplanetary trajectories that use gravity assists planetary flybys that increase spacecraft velocity without using onboard fuel. These trajectories allow Europa Clipper to gradually build energy while reducing the mission's propulsion demands.

The primary gravity-assist options considered included:

- VEEGA: Venus–Earth–Earth Gravity Assist
- EVEEGA: Earth–Venus–Earth–Earth Gravity Assist.
- MEGA: Mars–Earth Gravity Assist

While these paths extend the total flight time to approximately 5.5 to 6 years, they dramatically reduce the required characteristic energy (C3) from ~82 km²/s² (for a direct route) to about 15–20 km²/s². This reduction makes the mission compatible with commercially available launch vehicles like the Falcon Heavy, rather than relying on NASA's limited and costly SLS *(Buffington, B., 2014).*

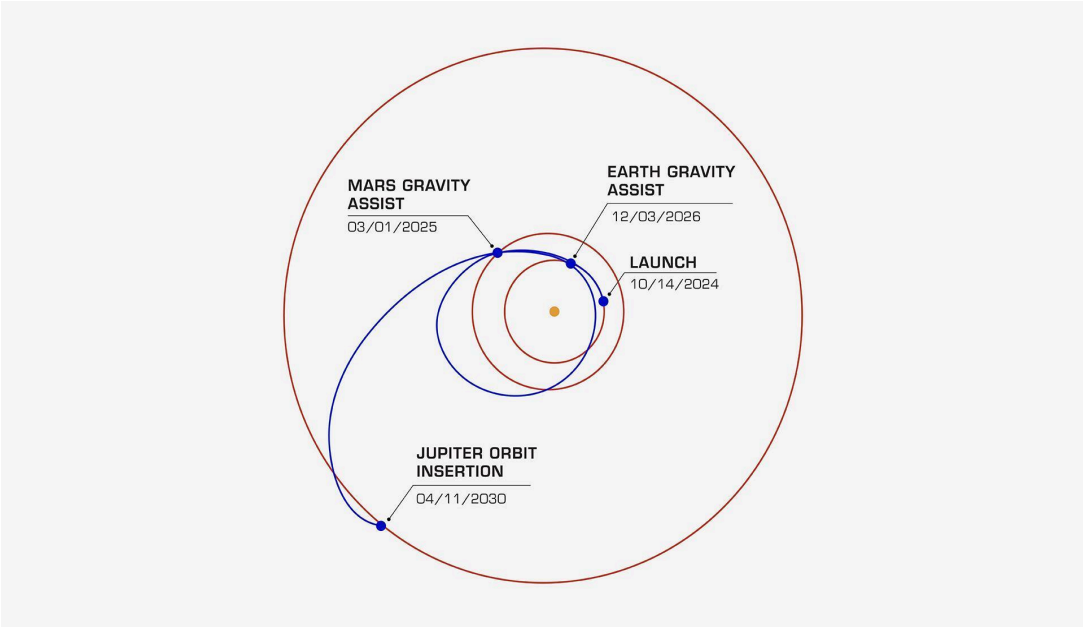

Figure 1. *MEGA trajectory for Europa Clipper showing launch (October 14, 2024), Mars and Earth flybys, and final approach to Jupiter with Ganymede assist before Jupiter Orbit Insertion (JOI) (NASA 2024).*

The mission team ultimately selected the MEGA trajectory for its balance of performance, safety, and flexibility. It avoids the thermal risks of Venus flybys, supports a more favorable mass margin, and allows for additional shielding and scientific payload *(Buffington et al., 2017).* Upon arrival at Jupiter, the spacecraft performs a Ganymede flyby to slow its approach and reduce the delta-V needed for orbit insertion, thus enhancing fuel efficiency and improving overall mission viability.

2.3. Gravity Assist: Physics and Function

A gravity assist (or slingshot maneuver) utilizes the gravitational field of a planet to alter a spacecraft's velocity and direction in the heliocentric (Sun-centered) frame. While the spacecraft appears to simply change direction relative to the planet, in the Sun's frame, it gains (or occasionally loses) velocity by effectively "borrowing" momentum from the planet's orbital motion*(Lam et al., 2015*),

Key parameters influencing the velocity change (Δv) include:
- The planet's orbital velocity
- The flyby angle (turn angle)
- The spacecraft's hyperbolic excess speed ($V\infty$)

Crucially, gravity assists require no onboard propellant, making them ideal for long-duration missions such as Europa Clipper. These maneuvers enable precise speed and trajectory adjustments that would otherwise demand infeasible fuel reserves. Additionally, gravity assists can alter the spacecraft's orbital inclination, arrival geometry, and timing, offering flexibility in mission planning and execution *(Buffington et al., 2017).*

2.4. Flyby Execution in Europa Clipper Mission

The mission architecture relies on a sequence of planetary and moon flybys to both achieve Jupiter orbit and enable rich scientific observation with minimal radiation exposure.

- Mars Flyby (Feb 2025): Alters spacecraft's trajectory and speed using Mars' gravity ($V\infty \approx$ 10.16 km/s).
- Earth Flyby 1 (Dec 2026): Major heliocentric speed increase (closest approach ~3,100 km).
- Earth Flyby 2 (Jan 2029): Refines final approach geometry and timing toward Jupiter.

On 11 April 2030, Clipper will reach Jupiter. A strategically timed flyby of Ganymede just before the JOI burn reduces Clipper's arrival velocity, minimizing fuel use. The spacecraft is then inserted into a 203-day orbit, setting the stage for repeated moon flybys*(NASA,2024)*.

Europa Clipper avoids direct Europa orbiting due to radiation intensity. Instead, it performs over 50 targeted flybys of Europa, as well as maneuvering via Ganymede and Callisto to refine trajectories. These operations allow:

- Minimal radiation exposure
- Effective data downlink intervals
- Adaptive trajectory design

Each Europa flyby (at altitudes as low as 25 km) supports core mission goals such as high-resolution imaging, subsurface radar mapping (via the REASON[1] instrument), surface composition analysis, and plume detection (*Buffington et al., 2017; Lam et al., 2015*),

[1]**REASON** is a radar instrument on Europa Clipper used to probe the subsurface structure of Europa's ice shell and search for subsurface water reservoirs.

2.5. Delta-V Maneuver Budget and Timeline

The interplanetary cruise phase of the Europa Clipper mission requires a series of deterministic delta-V maneuvers in addition to planetary gravity assists. These small but critical burns fine-tune the spacecraft's trajectory at key points in the mission.

The image below illustrates the full sequence of delta-V events, including:

- Trajectory Correction Maneuvers (TCM)
- Mars and Earth Gravity Assist targeting and cleanup burns
- Jupiter Orbit Insertion demonstration and main burn

Each maneuver is precisely timed to ensure optimal encounter geometry and minimal fuel expenditure.

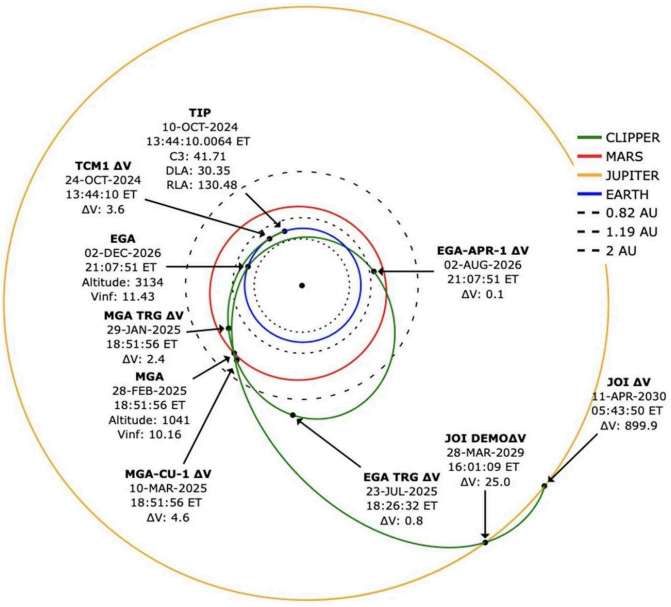

Figure 2. *Delta-V maneuver map along the Europa Clipper MEGA trajectory, with labeled burns and gravity assist events through 2030 (NASA 2024)*

These deterministic maneuvers total ~936.4 m/s, but gravity assists save over 5,000 m/s of ΔV, making the mission both fuel efficient and feasible (*Buffington et al., 2017*).

Part 3: Orbit Selection and Radiation Challenges

3.1 Radiation Environment at Europa

Europa resides inside Jupiter's intense magnetosphere, which is a significant hazard for spacecraft due to high-energy charged particles in this region. The primary contributors to this radiation are energetic electrons (> 1 MeV) and protons (>10 MeV). These data come from the GIRE (Galileo Interim Radiation Electron) and the DG83 (Divine-Garrett 1983) models, built based on Galileo mission data (*Lam et al., 2015*).

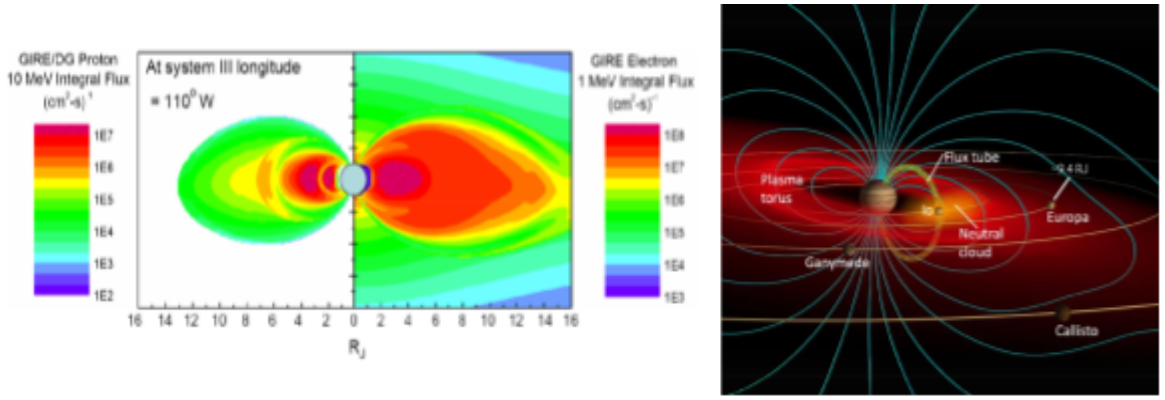

Figure 3. Contour plots of the GIRE/DG83 model E > 10 MeV integral proton (left) and E > 1 MeV electron (right) fluxes for the Jovian magnetosphere radiation re gion. The radiation environment at Jupiter is the strongest of any planet in the Solar System, and Europa's orbit is contained within a region of high radiation (96RJ). GIRE=Galileo Interim Radiation Electron model;9 DG83 = Divine and Garrett 1983 model.10 Figure courtesy of H. Garrett.

Europa orbits ~9.4 R_J from Jupiter. Here, radiation fluxes are sufficiently high that continuous exposure would rapidly exceed the survivability thresholds of spacecraft components. Without mitigation strategies, the expected Total Ionizing Dose (TID) would be multiple megarads, resulting in significant degradation and failure of onboard systems. The shielding factor depends strongly on materials and thickness. Europa Clipper's radiation vault uses materials like aluminum, titanium, and tantalum for shielding, which is estimated to reduce internal TID by exceeding 800 compared to unshielded exposure, maintaining cumulative mission exposure below the design threshold of approximately 3.14 Mrad (*Buffington et al., 2017; Lam et al., 2015*).

3.2 Orbit Selection Strategy for Radiation Mitigation

The Europa Clipper mission avoids remaining in orbit around Europa to save onboard electronics from the harsh radiation environment. Instead, the spacecraft is placed in a

Jupiter-centric orbit and performs around 49 targeted flybys of Europa over 3.5 years (*Buffington et al., 2017*).

This flyby strategy drastically limits cumulative radiation exposure by minimizing time spent in Europa's high-radiation zone. Each flyby dips briefly into the radiation belts, collects scientific data, and then the spacecraft moves back to safer regions further from Jupiter (*Lam et al., 2015*).

Critical trajectory techniques applied include:

- **Crank-Over-the-Top (COT) Sequences:** Inclination changes up to 20.1° via sequential Europa flybys to maximize global coverage while minimizing radiation accumulation.
- **Petal Rotation:** Gradual longitudinal shifting of flyby closest approach points to optimize coverage.
- **Switch-Flip Maneuver:** Transition between anti-Jovian and sub-Jovian hemispheres to exploit better illumination and radiation conditions.

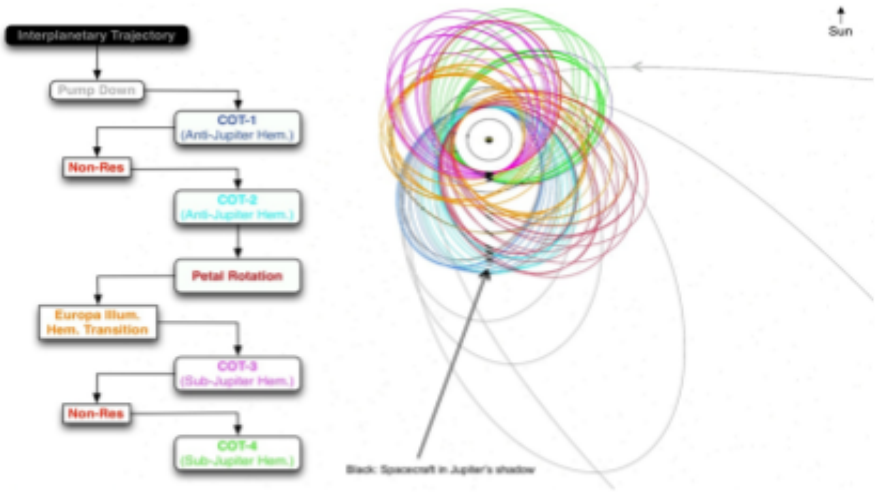

Figure 4: Viewed from Jupiter's northern hemisphere, centered at Jupiter in the rotating frame with the y-axis points to the Sun, z-axis normal to the Jupiter orbit plane, and x-axis completing the right rule. Gray: pump-down; blue: COT-1; cyan: COT-2; maroon: petal rotation; orange: Europa illuminated hemi sphere transition; magenta: COT-3; green COT-4; black: orbits of the four Galilean satellites. Spacecraft's trajectory shaded black when in Jupiter's shadow.

3.3 Radiation Exposure Management Across Mission Phases

Radiation exposure is managed across different mission phases. During the anti-Jovian hemisphere flybys (approximately 20 close approaches), the spacecraft accumulates a significant portion of its TID while conducting Crank-Over-the-Top (COT)

maneuvers. Following this, the Switch-Flip maneuver, using a Callisto flyby, transitions the orbit to the sub-Jovian hemisphere, where another ~20 flybys occur under improved illumination. Petal Rotation techniques are applied during both hemispheres to adjust the longitudinal spacing of flybys. The cumulative total ionizing dose across all phases is estimated between 2.8 and 3.14 Mrad, assuming the spacecraft's radiation vault shielding.

3.4 Remaining Radiation Challenges

Despite this well-designed trajectory, spacecraft still face cumulative radiation, which is a long-term risk for onboard electronics. Over time, the constant bombardment by energetic particles can lead to gradual degradation of electronic components, despite the use of advanced shielding and radiation-hardened systems. Additionally, sudden solar energetic particle (SEP) events could pose acute threats during sensitive flyby operations. The spacecraft includes shielding, redundant subsystems, and fault-protection software to manage these risks. Also, science operations are scheduled to avoid periods of peak radiation using predictive models of Jovian and solar activity.

Part 4: Navigation and Calculations

4.1 Trajectory Phases and ΔV Budget

Given the scientific importance of Europa first observed during the Galileo mission of the 1990's and also the previously mentioned challenges of addressing radiation concerns of Jupiter's magnetosphere, designing the trajectory for Europa Clipper is not an easy task, but an endeavor worth pursuing. In the early 2000s, NASA initially planned and canceled a mission known as the Jupiter Europa Orbiter, after the success of the Galileo mission to further observe Europa, Io, and Jupiter's magnetic atmosphere. It was canceled because orbiting directly in the magnetosphere would not be economically feasible (*Cangahuala, 2025*).

Instead, Europa Clipper and its flyby mission approach was selected and work began on the mission in 2011. The mission consists of five phases: Launch, Interplanetary Cruise, Arrival, Tour, and Disposal. Currently the mission is in the Interplanetary Cruise phase, in the subsection known as the Inner Cruise leg, flying from Mars back to Earth before heading out to Jupiter on the Outer Cruise Leg. Once completed, the Arrival phase will begin which includes the crucial Jupiter Orbit Insertion maneuver that will help ensure the success of the Tour and Disposal mission phases. The majority of fuel for Europa Clipper is reserved for the

JOI's maneuver and the trajectory for the flybys rely mostly on gravity assist maneuvers for the Tour phase.

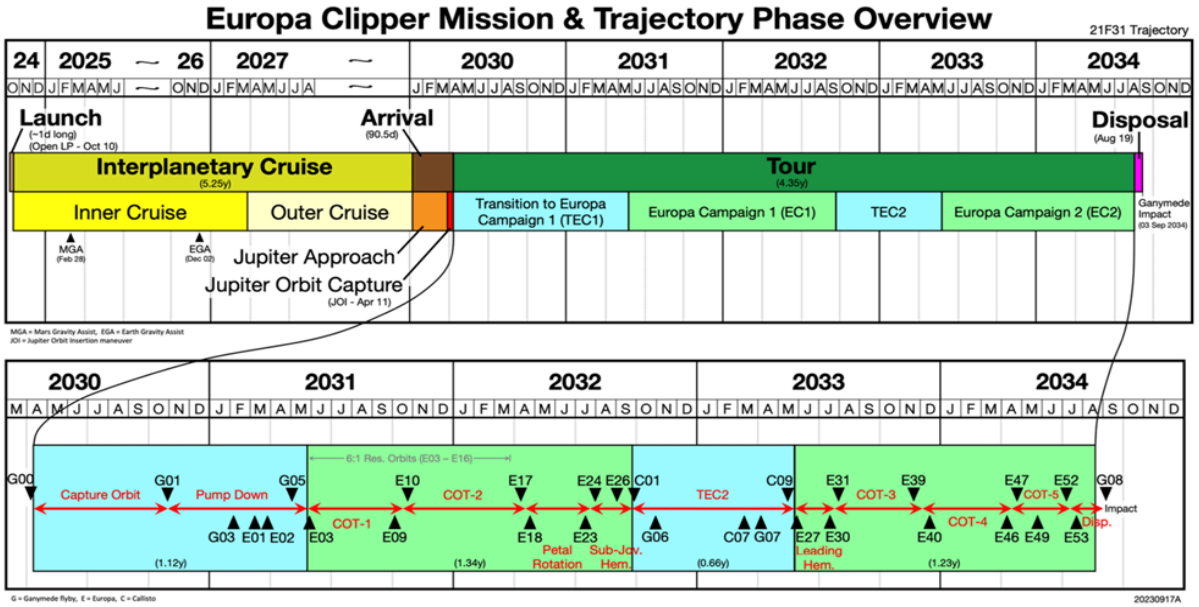

Figure 5. *Europa Clipper Mission and Trajectory Phase Overview with dates and maneuvers labeled (Cangahuala, 2025)*

Once Europa Clipper arrives into the Jovian system, by definition set by the mission parameters, with its altitude 200 km above Ganymede, the Tour phase will begin. The JOI maneuver begins 12 hours after reaching Ganymede and will have a burn that will last 6.1 hours. The orbit of this burn will be centered on Jupiter with an initial periapsis of 11.27 R_J. Which will then be raised at a velocity of about 53 m/s to perform the first scheduled flyby known as Ganymede-1 or G01 and the Tour pump-down will begin.

Depending on the success of JOI, whether there was overburn or underburn, Europa Clipper underutilized enough ΔV to allow up to two hours of delay in the mission before the Tour trajectory would have to be re-evaluated for the remainder of the mission. The first component of the Tour phase will be the Transition to Europa Campaign 1, a pump-down sequence of flybys designed to move the V∞ of 9.12 km/s down to 4.51 km/s to begin the low altitude flybys of Europa (*Cangahuala, 2025*). Europa Campaign 1 will focus on flybys as low as 25 km to survey the anti-Jupiter side of Europa. Europa is tidally locked to Jupiter and is in Laplace resonance with Io and Ganymede, which factored in the design for Crank-Over-the-Top maneuvers for the Tour phase.

After Europa Campaign 1, Transition to Europa Campaign 2 will occur. The reason for the transition between the two campaigns is the sunsetting on the anti-Jovian side of Europa and its rise on the sub-Jovian side during Europa Campaign 2. The altitudes and coverage of Europa's sub-Jovian side will be similar to its anti-Jovian coverage during Europa Campaign 1. The duration of both campaigns will be approximately 15 months and consist of 26 Europa flybys.

The Tour phase will conclude at the end of Europa Campaign 2 and will conclude with a Transfer maneuver to begin the disposal phase, where on 3rd of September, 2034, Europa Clipper will impact into Ganymede. This is a requirement for the decommission of Europa Clipper. The Tour phase is budgeted for a maximum use of ΔV of only 50 m/s or enough for one reverse crank maneuver. This was to ensure that in the case of loss of control of Europa Clipper during the Tour phase it will collide with Ganymede with a high probability. This was a mission requirement to prevent biological contamination of Europa that could occur if control was lost during the Tour phase and it collided with Europa. Meaning the trajectory of the Tour phase is almost exclusively gravity assist maneuvers with minimal ΔV input. In total, the required Δv for the mission is budgeted to be 1382 m/s with the majority (1007 m/s) reserved for JOI (*Cangahuala, 2025*).

4.2 Navigation Requirements

Constraints based on radiation considerations and fuel budget are not the only factors that went into the calculated trajectory for Europa Clipper. Science missions and the limits of the instruments on Europa Clipper played a role as well. The primary goal of navigation during the Interplanetary Cruise and Arrival phase was to reach the best conditions to perform maneuver G00 or the JOI under optimal conditions. Since the budgeted ΔV for JOI is the majority for the entire mission, it illustrates the importance of this one maneuver.

JOI was determined by the science, safety, and operability goals of the mission. The current operating strategy, with the trajectory model known as 21F31_V6 satisfies all requirements for the mission. For the science requirements, more constraints drive the trajectory. For example, for all flybys, it is required to have a baseline probability of impact (POI) of less than 5%. This alone is a hard constraint due to the LaPlace resonance and orbital structure of the trajectory as previously mentioned.

Other requirements that need specific constraints include the uncertainty in velocity when using the REASON VHF instrument. When below an altitude of 400 km and while the REASON VHF is in use, the velocity uncertainty can not be more than 44 mm/sec. Also, the

MISE instrument while collecting regional scale measurements can not have an altitude error of 0.375 km. REASON requires the velocity of uncertainty to be less than 5 m/s at the time of the last velocity measurement. Even line of sight measurements between Earth and Europa Clipper have a variance of only 50 km over the average span of roughly 5 AU. These factors lead as the main constraints from the scientific requirements for navigation accuracy.

Conclusion

Europa Clipper is a mission designed to maximize scientific measurements of a vital astrobiological target while navigating the unique challenges of operating within the Jupiter system. The scientific rewards outweigh the mission's expenses and were deemed essential for further exploration of Europa. To make the most of the constraints, the spacecraft emphasizes minimizing ΔV requirements while simultaneously maximizing coverage of Europa in a way that reduces overall radiation exposure. Instead of orbiting Europa directly, Europa Clipper's trajectory is centered around Jupiter, an approach critical for extending the mission's operational lifespan from weeks to years. The lion's share of the budgeted ΔV is dedicated to the Jupiter Orbit Insertion (JOI) maneuver, whose precision is paramount, as it directly impacts all subsequent gravity-assisted flybys during the Tour phase. A successful JOI is critical, with its execution determining the mission's overall success.

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